Saussure in HLLSet Algebra

Part V: Questions of Retrospective Linguistics

A Computational Exegesis of the *Course in General Linguistics*, Part Five, in the Boolean Lattice and GF(2) Ring of HLLSet Algebra

Fifth paper in the series *Saussure in HLLSet Algebra*

Alex Mylnikov Alexandr Mylnikov Jr. Deependra Kumar

October 19, 2026

Abstract

Part V closes the series by re-founding Part Five of the *Course*, *Questions of Retrospective Linguistics*, as time travel through old frames. The two perspectives of diachronic linguistics become the two directions of the commit DAG: the prospective reading follows the pivots forward, the retrospective reading inverts the transport matrices backward. The inversion is exact on the old span—the map $c \mapsto c\mathbf{M}$ is injective—and partial elsewhere, where the spill against the earliest frame measures the error of reading the present through the past's lens. The search for the earliest language and the prototype is re-founded as a fact about perishable origins: the lineage is not monotone, its frame transitions depart, retain, and add bits, and the first basis is an arbitrary origin that can be entirely replaced; the prototype is therefore a backward path along the lineage, not a projection into the first basis. Reconstruction becomes an executable inverse problem: the proto-node of a set of descendants is their meet, $\mathsf{P} = \bigcap_i X_i$, a view whose provenance points into the past; the reconstruction is exact exactly when nothing has departed, and the spill of P against the earliest frame measures what the descendants can no longer witness. The contribution of linguistic evidence to anthropology and prehistory is delimited with the same discipline: the DAG certifies shared ancestry of frames and shared nodes, and nothing else. Language families and linguistic types are then separated: a family is ancestry in the DAG, a type is similarity in the current fiber, and the algebra computes both without forcing them to agree. The paper, and the series, close with Saussure's own conclusion re-founded: the unique and true object of linguistics is langue envisaged in itself and for itself—and the lattice is exactly that object.

Keywords: Saussure; retrospective linguistics; prospective method; reconstruction; prototype; language family; linguistic type; time travel; old frames; transport inversion; meet; provenance; HLLSet; Boolean lattice; GF(2) ring; computational exegesis.

1 Introduction

1.1 Where Part V stands in the series

Part I [5] re-founded the general principles; Part II [6] the synchronic theory of value; Part III [7] the diachronic theory of the pivot; Part IV [8] the geographical theory of worlds and waves. Part Five of the *Course*, Questions of Retrospective Linguistics, is where Saussure turns the method around: instead of following the course of time, the linguist goes backward, reconstructing what the present can no longer show directly [2, Part V]. Part V of this series re-founds that backward method in the algebra, and with it closes the series.

1.2 Method, unchanged

The method is that of Parts I–IV [5, 6, 7, 8]: no metaphor, principles as theorems or invariants, ambiguity recorded as data. The translation note of Part I applies unchanged; references to the *Course* are by part, chapter, and section.

1.3 Roadmap

Section 2 fixes the standing definitions and adds the retrospective layer (prospective and retrospective readings, the reconstruction operator). Section 3 re-founds the two perspectives of diachronic linguistics as the two directions of the DAG. Section 4 re-founds the search for the earliest language and the prototype as a backward path along a perishable lineage, not a projection into the first basis. Section 5 is the center: reconstruction as the meet of descendants, with its exactness criterion. Section 6 delimits the anthropological evidence. Section 7 separates families from types. Section 8 closes the series with Saussure's own conclusion re-founded. Section 9 collects the dictionary of Part Five.

2 Preliminaries: the whole inheritance and the retrospective layer

We restate the standing definitions with their labels; proofs and motivation are in [5, 6, 7, 8].

Definition 2.1 (Standing definitions)**.** *The* ***plane*** *is* $\mathbb{B} = \{0, \dots, 32767\}$*; the* ***lattice*** *is* $\mathcal{B} = \mathcal{P}(\mathbb{B})$ *under* $\cup, \cap, \subseteq$ *and the Boolean ring* $\Delta = \Delta, \cap$*.* $\mu : \mathcal{T} \to \mathcal{B}$ *is an* ***IICA morphism****;* $\rho : \mathcal{B} \dashrightarrow \mathcal{C}$ *materializes through the door. A* ***basis frame*** $F = (B_1, \dots, B_k)$ *is a row-echelon GF(2) basis with distinct leading bits, stamped with a generation* γ*; hard keys, soft keys, and spill are as in Parts I–III. The* ***synchronic fiber*** *is* $\mathsf{Syn}(t) = (\mathcal{B}, F(t), \gamma(t))$*; a* ***pivot*** *is the extension/rotation of Part III; the* ***transport*** *across a pivot is* $c' = c\mathbf{M}$ *with the change-of-basis matrix* $\mathbf{M}$ *[7]. A* ***world*** *and a* ***bridge*** *are as in Part IV [8].*

2.1 The retrospective layer

Definition 2.2 (Prospective and retrospective readings)**.** *The* ***prospective reading*** *of a lineage is the forward traversal of the commit DAG: at each pivot, transport coordinates forward,* $c(t{+}1) = c(t)\mathbf{M}(t{+}1)$*. The* ***retrospective reading*** *is the backward traversal: given the present coordinates* $c(t)$*, recover the past coordinates* $c(t{-}1)$ *of any node that was already in the old span, and project nodes outside the old span onto the old frame with a measured error.*

Definition 2.3 (Reconstruction operator)**.** *Let* $X_1, \dots, X_n$ *be nodes (descendants) in the current lattice. The* ***reconstruction*** *of their common ancestor is the meet*

$$\mathsf{R}(X_1, \dots, X_n) \;=\; \mathsf{P} \;=\; \bigcap_{i=1}^{n} X_i,$$

read as a view*: a recipe whose sources are the descendants and whose provenance points into the past. The* ***reconstruction spill*** *is*

$$s(\mathsf{P}) \;=\; |\mathsf{P} \setminus \mathrm{cover}(F_0 \cap F_t)|$$

against the surviving *part of the earliest recorded frame* F_0*; the departed part* $F_0 \setminus F_t$ *is accounted as uncertainty (Definition 4.1), not as a lens.*

Proposition 2.4 (Backward reading is exact on the old span)**.** *Let* $F(t{-}1) = (B_1, \dots, B_k)$ *and* $F(t) = (B'_1, \dots, B'_{k'})$ *be two frames of a monotone ring,* $k' \geq k$*, with change-of-basis matrix* $\mathbf{M}$ *(*$k \times k'$*). Then:*

1. *the rows of* $\mathbf{M}$ *are the coordinates of the old basis in the new basis; they are linearly independent, so the map* $c \mapsto c\mathbf{M}$ *is injective;*
2. *consequently, for every node* $X \in \langle F(t-1)\rangle$*, the past coordinates are uniquely recoverable from the present coordinates: the equation* $c' = c\mathbf{M}$ *has at most one solution* c*, and it has one exactly when* c' *lies in the image (i.e. when* X *was in the old span);*
3. *for* $X \notin \langle F(t-1)\rangle$*, no past coordinates exist, and the old-frame projection has spill* $|X \setminus \mathrm{cover}(F(t-1))|$*, the measured error of reading the present through the past's lens.*

Proof. (1) The old basis vectors are linearly independent over GF(2), so their coordinate vectors are linearly independent; hence $\mathbf{M}$ has full row rank k, and left multiplication by $\mathbf{M}$ is injective. (2) Immediate from (1). (3) A node outside the old span has no old hard key; the spill is the part of it that the old frame cannot see, as in Part III [7, 12]. □

Remark 2.5 (Monotone span versus perishable frames)**.** *Proposition 2.4 states the retrospective method in the monotone-span regime of the EWM ring, where* $\langle F(t-1)\rangle \subseteq \langle F(t)\rangle$ *and no direction is ever lost. A real lineage is not monotone at the frame level: each new frame differs from its predecessor by a frame transition with departed, retained, and new parts, and the departed bits accumulate—at the limit they can replace the first frame entirely. In that regime the backward map is exact only on the retained part of the old span; the departed part is not inverted but reconstructed by hypothesis. This is the subject of Chapters II and III.*

3 Chapter I: The two perspectives of diachronic linguistics

Saussure distinguishes two ways of doing diachronic linguistics: the *prospective* method follows the course of time from past documents to the present; the *retrospective* method goes backward from the present to the past, and it is the method of reconstruction [2, Part V, ch. I].

Reading 3.1 (The two directions of the DAG). The prospective perspective is the forward traversal of the commit DAG: it sees the pivots as they happen, and transports coordinates forward. The retrospective perspective is the backward traversal: it sees the present fiber and asks which earlier fiber could have produced it, inverting transports by Proposition 2.4.

Observation 3.2 (The two methods have different exactness). The prospective reading is exact by construction: every pivot and every transport is recorded. The retrospective reading is exact only on the old span, and partial outside it, where the spill measures the error. Saussure's distinction between the two methods is therefore, in the algebra, a distinction between forward determinism and backward inversion with a measured residual—not a symmetrical pair of viewpoints.

4 Chapter II: Earliest languages and prototypes

Saussure's chapter II deflates the search for the first language: no living language is "older" than another in any absolute sense, and the prototype reconstructed by comparison is a hypothesis about relations, not a reality that was ever spoken [2, Part V, ch. II].

Definition 4.1 (Frame transitions along a lineage)**.** *Let* $F_0, F_1, \ldots, F_t$ *be the recorded frames of a lineage. Each step is a* ***frame transition*** *with departed, retained, and new parts*

$$\mathsf{Dep}_i = F_{i-1} \setminus F_i, \qquad \mathsf{Ret}_i = F_{i-1} \cap F_i, \qquad \mathsf{New}_i = F_i \setminus F_{i-1},$$

which partition $F_{i-1} \cup F_i$*. The* ***accumulated departure*** *from* F_0 *to* F_t *is*

$$\mathsf{Dep}(0,t) \;=\; F_0 \setminus F_t.$$

Nothing prevents $\mathsf{Dep}(0,t)$ *from growing until it equals* F_0*: the lineage is not monotone, and the first frame can be entirely replaced.*

Reading 4.2 (The first frame is an arbitrary and perishable origin). The earliest frame F_0 of a lineage is the first committed generation in the DAG. It is an *origin* only because the record starts there, not because the plane has an origin: by the coordinate-freedom theorem of Part I [5], any re-coordination of the plane is an automorphism, and no structural property prefers one first basis over another. And the origin is *perishable*: the lineage proceeds by frame transitions (Definition 4.1), whose departed parts accumulate. The prototype is therefore not a projection of the present into F_0—a lens that may have vanished—but a backward path along the lineage.

Observation 4.3 (No language is older than another). For every lineage, the first commit is arbitrary: it is the point where the record begins, and a different beginning would be an isomorphic coordinate choice. The question "which language is oldest" therefore has no algebraic meaning: all lineages are equally old, because age is a property of the record, not of the lattice. Saussure's deflation of the question is re-founded as the arbitrariness of the origin—an origin that, in addition, does not survive its own lineage.

Proposition 4.4 (The prototype is a backward path, not a projection)**.** *Let $F_0, \dots, F_t$ be the frames of a lineage and X a present node.*

1. *A single projection of X into F_0 is not the prototype: it reads the present through an origin that may have departed in part or in whole. When $\mathsf{Dep}(0,t) = F_0$, the lens F_0 is foreign to the present frame ($F_0 \cap F_t = \varnothing$), and the projection measures similarity to a vanished system, not descent from it.*
2. *The correct prototype reading of X is the backward path along the lineage: transport X's coordinates backward through each* retained *part Ret_i, and supply the* departed *parts Dep_i only by the comparative hypothesis of Section 5, applied now to oppositions rather than to nodes.*
3. *The accumulated departure $\mathsf{Dep}(0,t)$ is the exact uncertainty of the prototype: the part of the origin that the present can no longer witness, and that only hypothesis—never the record—can supply.*

5 Chapter III: Reconstructions

Saussure's chapter III is the heart of Part Five: reconstruction is comparative, the starred form is a hypothesis about *relations*, and its value lies in the correspondences it summarizes, not in any claim about substance [2, Part V, ch. III].

Reading 5.1 (Reconstruction is a meet). The comparative method places descendant forms side by side and asks what they have in common. In the lattice, the descendants are nodes $X_1, \dots, X_n$, and what they have in common is the largest node contained in all of them—the meet $\mathsf{P} = \bigcap_i X_i$. The starred form is P read as a view: a recipe over the descendants' provenance, stamped with the present generation, and explicitly *not* an original.

Proposition 5.2 (Exactness of reconstruction)**.** *Suppose the true ancestor A exists in the DAG and each descendant X_i was formed from A by joins (additions of bits), so $A \subseteq X_i$ for all i. Then:*

1. *$A \subseteq \mathsf{P}$: the reconstruction never claims less than the ancestor;*
2. *$\mathsf{P} \setminus A$ is the innovation shared by* all *descendants (the convergent addition), which reconstruction cannot distinguish from inheritance;*
3. *$A \setminus \mathsf{P}$ is the part of the ancestor that departed in at least one descendant, which the descendants alone cannot recover;*
4. *reconstruction is exact ($\mathsf{P} = A$) if and only if nothing departed, i.e. $A \setminus \mathsf{P} = \varnothing$.*

The reconstruction spill of Definition 2.3 measures the part of P that the surviving *part of the earliest frame cannot see; the departed part of the earliest frame is itself uncertainty. The differences $\mathsf{P} \setminus A$ and $A \setminus \mathsf{P}$ are the two directions of the reconstruction's uncertainty.*

Proof. (1) $A \subseteq X_i$ for all i implies $A \subseteq \bigcap_i X_i = \mathsf{P}$. (2) If a bit lies in every descendant but not in A, it was added convergently in all lines; the meet cannot tell addition from inheritance. (3) A bit of A missing from some descendant is missing from the meet. (4) Immediate from (1)–(3). □

Corollary 5.3 (Correspondences are transported coordinate agreements)**.** *When the descendants live in different worlds or different generations, the comparative method first transports their hard keys to a common reference frame (an ancestor frame if one is shared); a* correspondence *between two descendants is the equality of transported coordinates on a set of oppositions. The reconstruction* P *is then the node whose transported coordinates are the meet of the descendants' coordinates—the exact summary of the correspondences, and nothing more.*

Remark 5.4 (The starred form is a hypothesis, not a sound)**.** *Saussure insists that a reconstruction is a set of relations, not a claim about how the proto-language sounded. The algebra agrees precisely:* P *is a view—an expression over the descendants—and it is never stored as an original. Its generation stamp is the present generation in which the hypothesis is evaluated, and its provenance points into the past without claiming that the node itself was ever an observed original.*

6 Chapter IV: Linguistic evidence in anthropology and prehistory

Saussure's chapter IV asks what linguistics can contribute to anthropology and prehistory, and answers with characteristic discipline: linguistic evidence is structural, and it must not be made to prove what only other sciences can prove [2, Part V, ch. IV].

Reading 6.1 (The DAG certifies ancestry, nothing else). The algebra's evidence is of exactly three kinds: shared nodes (cognates), shared frames (a common system of oppositions), and shared pivots (a common history of changes). From these it can certify common ancestry of frames and the order of divergences. It cannot certify anything about the speakers' genes, pots, or migrations, because bits, generations, and provenance carry no such information.

Observation 6.2 (The discipline of the lattice). The lattice disciplines the inference in both directions. It *forces* the linguist to say exactly what is shared—a node, a frame, a pivot—and it *forbids* conclusions that the shared objects do not support. Saussure's caution about linguistic evidence is therefore not a stylistic preference; it is a type discipline: an inference is valid only if it is a function of the DAG, the worlds, and the provenance, and nothing else.

7 Chapter V: Language families and linguistic types

Saussure's last chapter separates two classifications that are constantly confused: the *family* is genealogical, the *type* is structural. They can and do diverge [2, Part V, ch. V].

Reading 7.1 (Families are ancestry; types are profiles). A family is a property of the DAG: two worlds belong to the same family when their frames descend from a common ancestor frame. A type is a property of the current fiber: two worlds are of the same type when their soft-key profiles against a reference frame are close. The two computations use disjoint data and need not agree.

Definition 7.2 (Family and type)**.** *The* ***family*** *of a world* W *is*

$$\mathsf{Fam}(\mathsf{W}) \;=\; \{\, \mathsf{W}' : F(\mathsf{W}) \text{ and } F(\mathsf{W}') \text{ descend from a common ancestor frame} \,\},$$

an ancestry relation in the frame DAG. The ***type*** *of a world against a reference frame F is its profile*

$$\mathsf{Typ}_F(\mathsf{W}) \;=\; \big(\mathsf{BSS}_{B_1}(\mathcal{N}(\mathsf{W})), \ldots, \mathsf{BSS}_{B_k}(\mathcal{N}(\mathsf{W}))\big),$$

a similarity relation in the current fiber. Family is a fact of provenance; type is a fact of the present.

Proposition 7.3 (Families and types are independent)**.** *Two unrelated worlds may be type-identical (convergence): their histories differ but their present profiles coincide. Two related worlds may be type-distant (divergence): their histories coincide up to a point but their profiles have drifted apart. The algebra computes both and never forces them to agree; Saussure's warning against confusing family and type is re-founded as the independence of the ancestry relation and the profile metric.*

Proof. Ancestry is determined by the DAG; type is determined by BSS against a fixed frame. The two functions have different domains (history vs present) and different data (provenance vs soft keys); nothing in the definitions couples them. □

8 Conclusion of the series: langue in itself and for itself

Saussure closes the *Course* with the sentence that defined a century: linguistics has for its unique and true object *la langue envisagée en elle-même et pour elle-même*—language studied in itself and for itself [2, Conclusion]. The series closes by re-founding that sentence, and it does so with an exactness that no philosophical commentary could offer.

In itself. The lattice $\mathcal{B}$ over the plane $\mathbb{B}$ is studied in itself: its Boolean structure and its GF(2) ring, its frames, generations, pivots, and transports. Nothing outside the lattice is needed to state its laws; the panchronic axioms of Part I are the exact form of "in itself."

For itself. The lattice is not the servant of any host, any vocabulary, any speaker. It is the system of differences that Parts I–V re-founded chapter by chapter: the sign as (μ, ρ), arbitrariness as coordinate freedom, value as the hard key, the pivot as the diachronic event, analogy as the fourth proportional, worlds as languages, bridges as waves, reconstruction as the meet. The algebra interprets the *Course*; the *Course* interprets the algebra; and the object of both is one and the same: langue, as a lattice, in itself and for itself.

9 The dictionary of Part Five

10 What Part V adds, and what the series as a whole adds

1. **Backward reading becomes a computation.** Proposition 2.4 gives the retrospective method its exact domain: injective inversion on the old span, measured spill outside it.
2. **Reconstruction becomes an executable inverse problem.** Definition 2.3 and Proposition 5.2 turn the starred form into a meet with a proven exactness criterion and a measured uncertainty in both directions ($\mathsf{P} \setminus A$ and $A \setminus \mathsf{P}$).
3. **The prototype is demystified.** The first basis is an arbitrary origin by Part I's coordinate freedom, and it is perishable: the lineage proceeds by frame transitions whose departed parts accumulate. The prototype is therefore a backward path along the lineage, with the accumulated departure as its uncertainty—not a projection into a vanished origin.
4. **Family and type are separated as datatypes.** Definition 7.2 and Proposition 7.3 make the genealogical/structural distinction a difference of data, not of emphasis.
5. **The series closes the circle.** The *Course* has been read out of the algebra in five parts; the algebra has been read out of the *Course* in the five EWM papers [10, 11, 12, 13, 14]; the two directions meet in one construction, and the closing object is Saussure's own: langue, in itself and for itself.

Part of the *Course*	In the *Course*	In HLLSet Algebra (this series)
Introduction	langue and parole; the object of linguistics	the morphism step and the lattice as langue [14]; folded into Part II
Part One	general principles	sign, arbitrariness, linearity, immutability/mutability, synchrony/diachrony [5]
Part Two	synchronic linguistics	value as the hard key; differences without positive terms; syntagmatic/associative readings [6]
Part Three	diachronic linguistics	the pivot; sound change, analogy, folk etymology, agglutination; diachronic units [7]
Part Four	geographical linguistics	worlds, codebooks, thresholds; diversity, continua, substrata, waves [8]
Part Five	retrospective linguistics; conclusion	time travel through old frames; reconstruction as the meet; families vs types; langue in itself and for itself [9]

Table 1: The complete architecture of the series: the *Course* in General Linguistics, re-founded part by part in HLLSet Algebra.

Acknowledgments

This paper was created in collaboration with, and with the assistance of, an AI system—DeepSeek (deepseek-v4-pro). The meet reading of reconstruction, the injectivity reading of the retrospective method, the family/type separation, and several of the formulations emerged in dialogue with the assistant, whose drafts the authors revised and take full responsibility for in their final form.

References

[1] F. de Saussure, *Cours de linguistique générale*, Payot, Paris, 1916. (Edited posthumously by C. Bally and A. Sechehaye, with A. Riedlinger.)

[2] F. de Saussure, *Course in General Linguistics*, translated and annotated by Roy Harris, Duckworth, London, 1983.

[3] F. de Saussure, *Course in General Linguistics*, translated by Wade Baskin, Philosophical Library, New York, 1959.

[4] F. de Sossyur, *Trudy po yazykoznaniyu* [Works in Linguistics], edited by A. A. Kholodovich, translation of the *Cours* by A. M. Sukhotin, Progress, Moscow, 1977. (Authors' working copy; used for verification of the readings.)

[5] A. Mylnikov, A. Mylnikov Jr., and D. Kumar, *Saussure in HLLSet Algebra, Part I: General Principles*, SAUSSURE PUB series, 2026.

[6] A. Mylnikov, A. Mylnikov Jr., and D. Kumar, *Saussure in HLLSet Algebra, Part II: Synchronic Linguistics*, SAUSSURE PUB series, 2026.

[7] A. Mylnikov, A. Mylnikov Jr., and D. Kumar, *Saussure in HLLSet Algebra, Part III: Diachronic Linguistics*, SAUSSURE PUB series, 2026.

Saussure (Part Five)	In the *Course*	In HLLSet Algebra
prospective method	follow the course of time	forward traversal of the DAG; transports $c' = c\mathbf{M}$: Definition 2.2
retrospective method	go backward from the present	backward inversion on the old span, projection with spill elsewhere: Proposition 2.4
earliest language	no language is older than another	the first commit is an arbitrary origin: Observation 4.3
prototype	a hypothesis, not a reality	a backward path along the lineage; the first basis is arbitrary and perishable, and $\mathsf{Dep}(0,t)$ is the uncertainty: Proposition 4.4
reconstruction	comparative; the starred form	the meet $\mathsf{P} = \bigcap_i X_i$, a view with past provenance: Definition 2.3, Proposition 5.2
correspondences	the relations reconstruction summarizes	transported coordinate agreements: Corollary 5.3
linguistic evidence in prehistory	structural, disciplined	shared nodes, frames, pivots certify ancestry, nothing else: Observation 6.2
language family	genealogical	ancestry in the frame DAG: Definition 7.2
linguistic type	structural similarity	soft-key profile in the current fiber: Definition 7.2
family vs type	they can diverge	independence of ancestry and profile: Proposition 7.3
the conclusion of the *Course*	langue in itself and for itself	the lattice in itself and for itself: Section 8

Table 2: The complete dictionary of Part Five of the *Course* into HLLSet Algebra.

[8] A. Mylnikov, A. Mylnikov Jr., and D. Kumar, *Saussure in HLLSet Algebra, Part IV: Geographical Linguistics*, SAUSSURE PUB series, 2026.

[9] A. Mylnikov, A. Mylnikov Jr., and D. Kumar, *Saussure in HLLSet Algebra, Part V: Questions of Retrospective Linguistics*, SAUSSURE PUB series, 2026.

[10] A. Mylnikov, A. Mylnikov Jr., and D. Kumar, *Emerging World Models: A Structural Framework for Self-Organizing World Representations*, EWM PUB series, 2026.

[11] A. Mylnikov, A. Mylnikov Jr., and D. Kumar, *The EWM Lattice and Its Worlds*, EWM PUB series, 2026.

[12] A. Mylnikov, A. Mylnikov Jr., and D. Kumar, *Transformations and Trajectories of a Content-Addressed State Machine*, EWM PUB series, 2026.

[13] A. Mylnikov, A. Mylnikov Jr., and D. Kumar, *Attention on Lattice Traversal Is All You Need*, EWM PUB series, 2026.

[14] A. Mylnikov, A. Mylnikov Jr., and D. Kumar, *Langue as a Lattice: Closing the Circle from Statistical to Structural Linguistics through Content-Addressed Algebras*, EWM PUB series, 2026.

[15] G. Birkhoff, *Lattice Theory*, American Mathematical Society, 1940.

[16] S. Mac Lane, *Categories for the Working Mathematician*, Springer, 1971.

[17] L. Hjelmslev, *Prolegomena to a Theory of Language*, University of Wisconsin Press, 1961.

[18] Z. S. Harris, *Distributional Structure*, Word, 10(2–3):146–162, 1954.

[19] J. R. Firth, *A Synopsis of Linguistic Theory, 1930–1955*, Studies in Linguistic Analysis, 1957.

[20] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, *Attention Is All You Need*, NeurIPS, 2017.